Plane and stereographic projections of convex polyhedra from minimal information

By A. J. Cole*

A method of using a digital plotter to draw plane and stereographic projections of convex polyhedra is described. The polyhedron may be regarded as either a framework or as an opaque or semi-transparent solid body. The input information consists of only the space coordinates of the vertices, the viewpoint and the centre of vision.

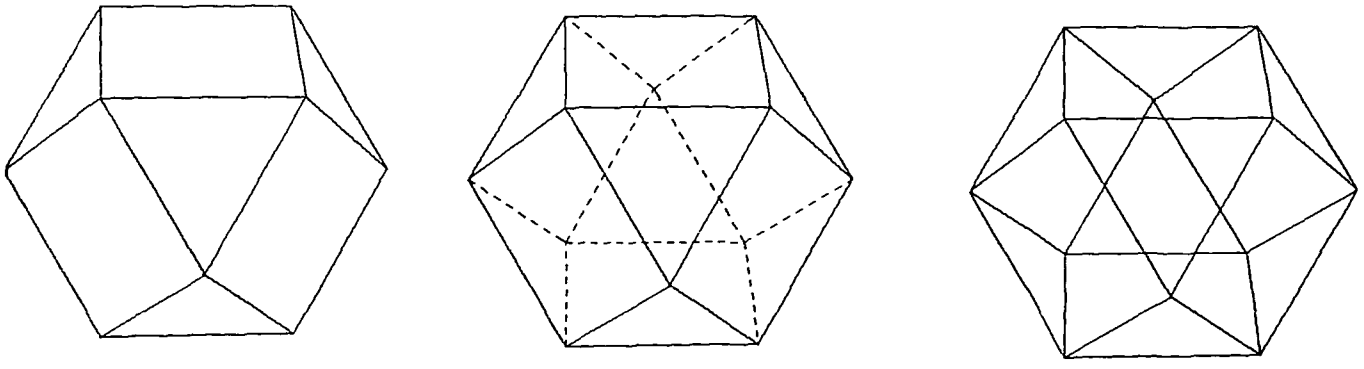
Many programs have been written, usually for engineering purposes, to draw plane, trimetric, isometric and stereographic projections of frameworks which are not necessarily convex. The word *framework* here is used to imply that the technique is not used to draw solid bodies in the sense that only those edges which are visible from the given viewpoint are drawn. The technique used is to specify, in addition to the viewpoint and centre of vision, both the space coordinates of the vertices and the way in which they are joined in pairs to form the rods of the framework. The vertices are then projected into a plane perpendicular to the line joining the viewpoint to the centre of vision by the method briefly described in the next section or by some similar method. The virtue of a particular program can then be judged by the efficiency with which the edges are drawn by the plotter. The resulting diagram is meaningful to a draughtsman but is frequently of little meaning to other viewers.

More recently G. O. Gellert (1965) has described some geometric computing techniques which can be applied to a number of design problems including that of drawing plane projections of various types. These methods are not described in detail but the text suggests that a large and powerful computer is required.

In this paper we describe a method of drawing plane and stereographic projections of convex polyhedra from minimal information. A convex polyhedron is completely determined by its vertices or alternatively by its bounding planes. We consider the case when the vertices are specified since it is comparatively easy to find the vertices when the planes are given. The edges which are visible from the viewpoint are then determined and their plane projections drawn. This part of the projected figure is shown for a cubeoctahedron in **Fig. 1(*a*)** and is an example of an opaque polyhedron. The edges which are not visible can then be determined and may be either drawn dotted or solid as shown in **Figs. 1(*b*)** and **1(*c*)**, respectively. By drawing two such projections from adjacent viewpoints it is now possible to produce stereographic effects as shown in **Fig. 2** for a truncated cube. It is also possible to shade the faces with a shading density proportional to the angle between the normal to the plane of the face and the direction of the source of illumination. An example of this type of shading is shown in **Fig. 3**.

The method does not depend on any symmetries of the body. For example in **Figs. 4(*a*)**, **4(*b*)**, **4(*c*)** we show projections of a solid, semi-transparent and transparent polyhedron with vertices chosen at random on the surface of a sphere.

The method in its full generality does not extend to



1(a) 1(b) 1(c)

Fig. 1.—Opaque, semi-transparent and transparent cube octahedron

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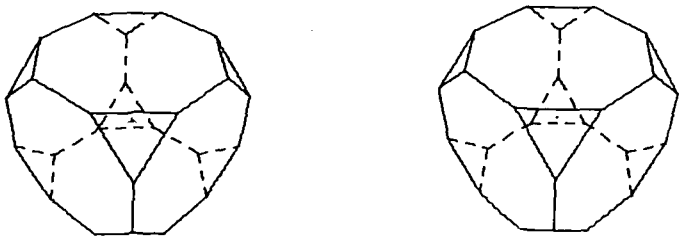


Fig. 2.—Stereographic effect for a truncated cube

non-convex bodies since these are not in general uniquely specified by their vertices. It is easy to see this by considering a tetrahedron with one internal vertex. There are then four simple but non-convex polyhedral bodies determined by this set of five vertices, obtained in effect by pressing in any one of the faces of the original tetrahedron. Further, a solid body which is not simply connected, and this occurs frequently in modern architecture, is *not* in general uniquely specified by its vertices and edges together.

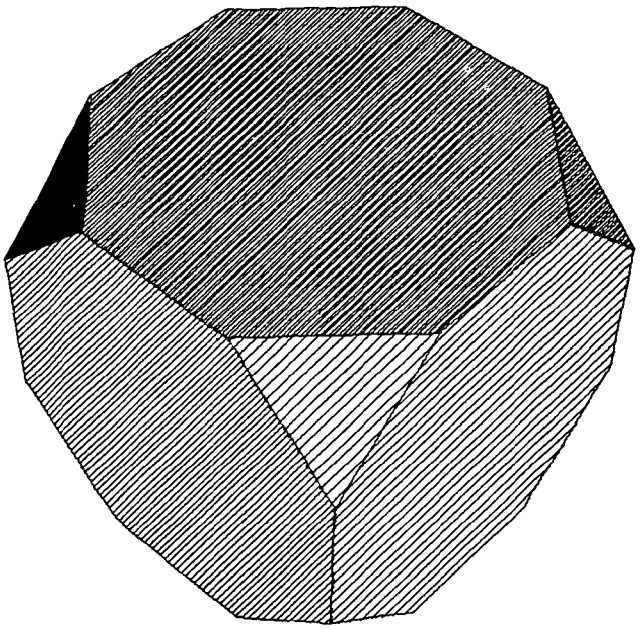


Fig. 3.—Illuminated polyhedron

The basic projection

Let $V(p, q, r)$ and $C(u, v, w)$ be the viewpoint and centre of vision respectively. Then VC has direction cosines

$$(l, m, n) = ((u - p)/k, \quad (v - q)/k, \quad (w - r)/k)$$

where

$$k^2 = (u - p)^2 + (v - q)^2 + (w - r)^2,$$

and by a simple change of coordinates

$$x' = x - p, \quad y' = y - q, \quad z' = z - r$$

we transform the origin to V. The plane π through the midpoint of VC with VC as normal now has equation

$$x'(u - p) + y'(v - q) + z'(w - z) = \tfrac{1}{2}((u - p)^2 + (v - q)^2 + (w - r)^2)$$

and so a vertex of the original body is transformed into a point

$$((a - p)t, \quad (b - q)t, \quad (c - r)t)$$

of π where

$$t = \frac{\tfrac{1}{2}((u - p)^2 + (v - q)^2 + (w - r)^2)}{(a - p)(u - p) + (b - q)(v - q) + (c - r)(w - r)}$$

The rotation matrix

$$\begin{vmatrix} -(v - q)/\lambda & (u - p)/\lambda & 0 \\ -(u - p)(w - r)/\lambda & -(v - q)(w - r)/\lambda & \lambda \\ u - p & v - q & w - r \end{vmatrix}$$

where $\lambda^2 = (u - p)^2 + (v - q)^2$, now transforms VC

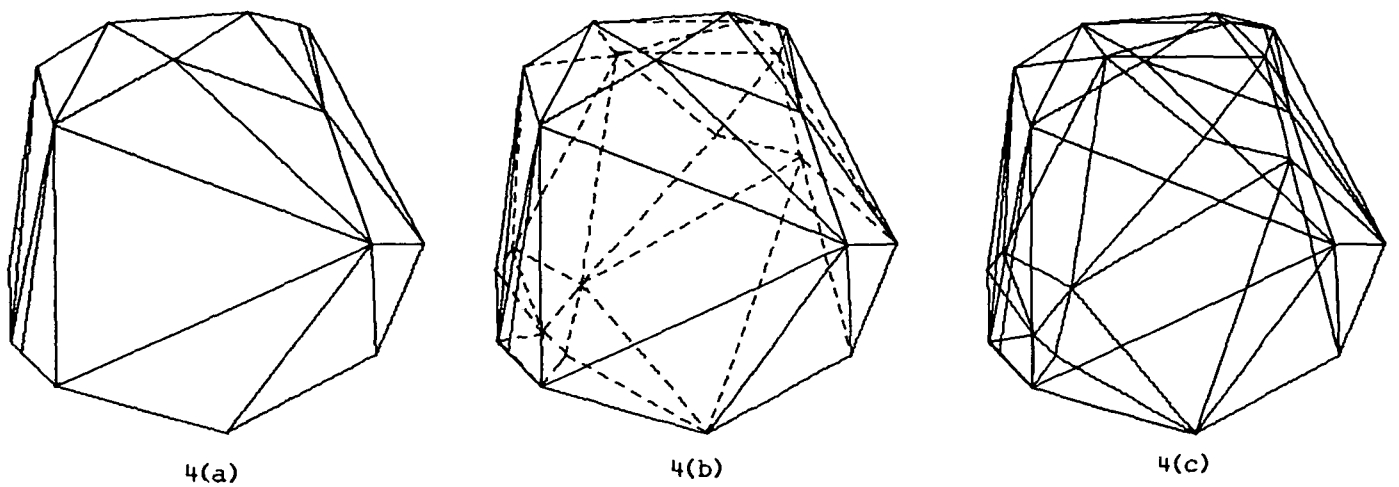


Fig. 4.—Polyhedron with vertices at random on the surface of a sphere

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into the new z-axis and so transforms the space coordinates (a, b, c) of an original vertex into plane coordinates in π. The scale factor can be ignored since another scale factor has to be applied at the digital plotting stage.

For the remainder of this paper we will refer to space coordinates by lower-case triples such as (x, y, z), and the corresponding plane coordinates by upper-case pairs such as (X, Y), or briefly as p and P, respectively.

The general method

In this section we describe briefly the complete method for determining the visible and non-visible edges, and then consider the various steps in more detail in subsequent paragraphs. One difficulty at first sight is in knowing just where to start with a vertex which is visible from the viewpoint. This is not necessarily the nearest vertex since this can be on the opposite side of a large bounding plane. It is, however, true that some space vertices corresponding to vertices on the perimeter of the convex closure of the projected vertices are visible, and one of these is simply a point of the projected set with minimum Y coordinate—and if there are several such, that with minimum X coordinate within this subset. This may still not uniquely specify a single point, as can be seen by considering a viewpoint on the extension of a side of a cube, but in this case, reverting to the space coordinates, that one nearest to the viewpoint can certainly be seen.

With a starting point now available it is possible to determine the successive points on the projected perimeter $P_1P_2P_3 \ldots$ by at the ith stage finding the vertex P_{i+1} such that the join P_iP_{i+1} makes a minimum angle with the previous join $P_{i-1}P_i$, care being taken to reject vertices which cannot be seen. Suppose there are k such projected vertices $P_1P_2 \ldots P_k$. Then P_kP_1 is the edge which closes the corresponding polygon. Every side of the polygon can be seen and so it can be drawn.

The basic idea for finding the other visible edges is as follows. Take any two adjacent vertices P_jP_{j+1} on the perimeter and find the nearest plane to the viewpoint through the two corresponding space vertices P_iP_{i+1} and some other space vertex P_t. Find all other space vertices in this plane which may include some more on the previously found perimeter. This set of vertices will form the bounding edges of a visible plane face of the polyhedron whose plane projection can now be drawn.

If for certain technical reasons described later this is not desirable then we abandon this approach and repeat the process with a new adjacent pair of vertices on the perimeter. Otherwise, the new edges are drawn and the previous perimeter is modified to exclude some of its original vertices and to include the new ones. The process terminates when all coordinates of the current perimeter lie in the nearest bounding plane to the viewpoint.

The non-visible edges of a solid polyhedron can now be found by an analogous method by using planes farthest from the viewpoint instead of nearest.

Some technical considerations

Some slight technical difficulties arise from the round-off error in the course of computation, and also from the fact that the lines drawn by the digital plotter have a finite thickness.

Round-off error can cause points not to exactly satisfy the equations of planes or lines in or on which they should lie. Similarly it can happen by unfortunate round-off that the point with minimum Y coordinate found to initiate the method may in fact not be visible. Both of these difficulties are simply overcome by using a pair of inequalities instead of a single equality, and since this introduces only trivial but tedious program modifications we will ignore them for the rest of this paper.

In a similar way two adjacent edges may in fact coincide when plotted, although theoretically they are distinct. There is nothing much lost in plotting both lines and we will again ignore this technicality.

The initial bounding perimeter

As described in the general method the first step is to find the projected point with minimum Y coordinate, and if this is not unique to find the point with minimum X coordinate in this subset which is nearest to the viewpoint. Call this point P_1. The corresponding point p_1 is a visible point and P_1 is on the perimeter of the projected set. The line of minimum slope amongst those joining P_1 to all other points of the projected set is now found. There may be several points $Q_1, Q_2, \ldots Q_r$ of the set on this line, not all of which are necessarily visible. Find next the point Q_1 from amongst these which is firstly farthest from P_1; if there are several such equidistant from P_1 then find the one whose corresponding spatial vertex is nearest to the viewpoint. This is also visible and so are those points from amongst $q_1, q_2, \ldots q_r$ which lie on p_1q_j. These are now ordered by the distance from P_1 of their plane equivalents, giving an initial set $P_1, P_2, \ldots P_i$ of successive points on the perimeter of the plane set. The process continues using P_i as the new starting point and with the same method as above excepting that the line of minimum slope is replaced by the line making the smallest positive angle with $P_{i-1}P_i$. It terminates when the last point P_{k+1} just found coincides with P_1. The set $P_1P_2 \ldots P_k$ is the required bounding polygon.

The remaining visible edges

To save time in subsequent calculation the points $p_1p_2, \ldots p_k$ can now be deleted from the original vertex set, although a record of them and of $P_1, P_2, \ldots P_k$ must be kept in the current perimeter set. The plane through p_1p_2 and another vertex p_r of either the remaining spatial vertices or one of $p_3, p_4, \ldots p_k$ which is nearest to the viewpoint V is now found. This is most simply done vectorially by choosing one at random and finding the corresponding plane $ax + by + cz + d = 0$. Now work systematically through the other points

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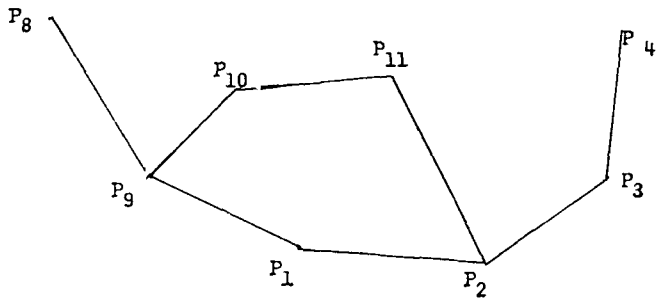


Fig. 5.—Modification of the polygonal boundary

evaluating $t = |d/(ax + by + cz)|$. If for p_s, $t \geqslant 1$ then proceed to the next point. If $t < 1$ then replace p_r by p_s, find the new plane through $p_1p_2p_s$ and continue. The final plane found by this method is now visible and so also are any other vertices lying in it. These can now be simply ordered in the plane of projection, and those edges which have already been drawn can be output on the plotter. The parameters a, b, c, d are recorded for subsequent use, and the plane perimeter $p_1p_2 \ldots p_k$ is modified to exclude those edges which are finished with and to include those which are new. A typical case is shown in **Fig. 5** where $P_1P_2 \ldots P_9$ was the original polygon and $P_1P_2P_{11}P_{10}P_9$ are points in the nearest plane. The new perimeter will be $P_2P_3 \ldots P_9P_{10}P_{11}$, or, better from the point of view of subsequent minimal pen movement, the ordered set $P_{11}P_2P_3 \ldots P_9P_{10}$.

The process now continues taking $P_{11}P_2$ as the new initial line, and a new nearest plane

$$ax + by + cz + d = 0$$

is found. This will frequently but not always contain P_3. There are now two cases in which this plane is rejected. The first is when it has already occurred, which can be found by checking against the recorded list of previous values of a, b, c, d. An example is shown in **Fig. 6** in which part of a cube has already been drawn and in

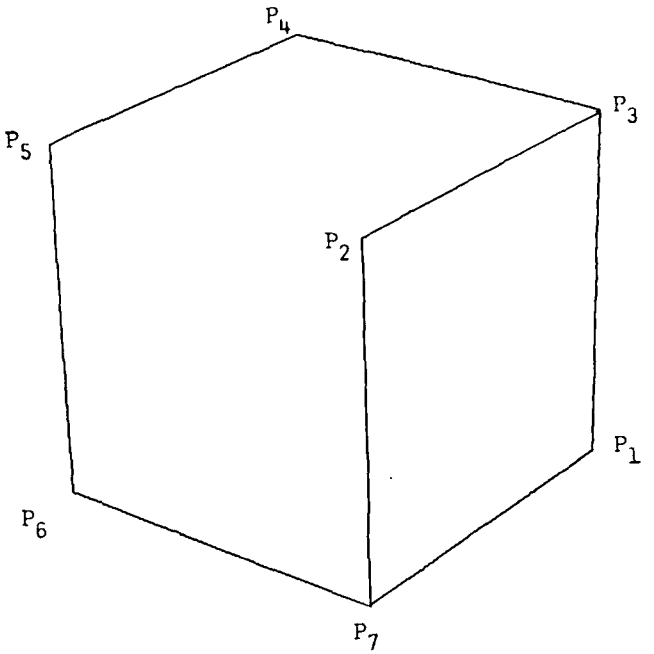
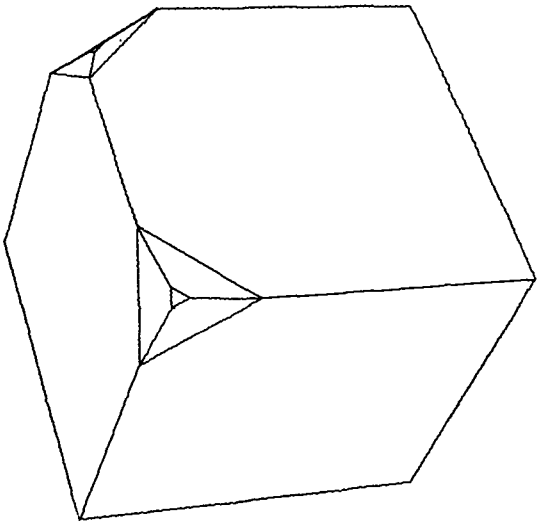


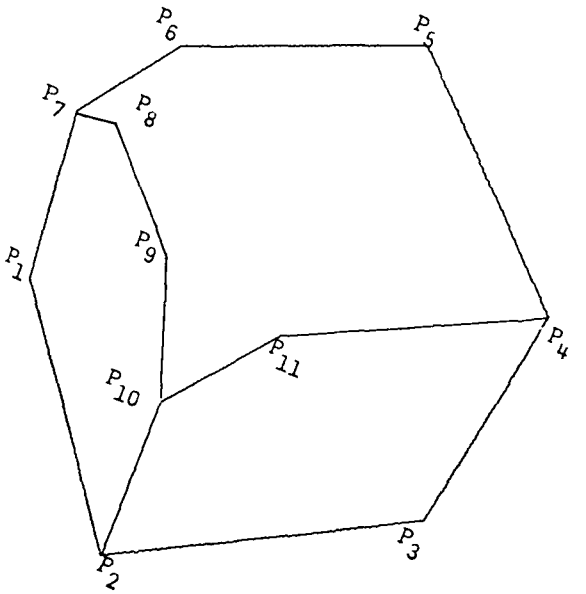
Fig. 6.—First case of rejection of a new plane

which, if $P_7P_2P_3P_4P_5P_6$ is the new boundary and by mischance the current plane is found to be $P_7P_2P_1$, then this has already been accounted for. This example is trivial and could easily be treated differently, but more complicated and less trivial cases occur in practice.

The second case to be rejected is when the plane just found divides the remaining perimeter into two disjoint pieces. This is illustrated in **Fig. 7(*b*)** where the figure to be drawn is a cube with two adjacent corners chopped off and some structure such as a tetrahedron raised on the



7(a)



7(b)

Fig. 7.—Second case of rejection of a new plane

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two faces so exposed. The full figure required is shown in **Fig. 7(*a*)**. After drawing the perimeter as described above and starting from P_1 the first face to be found will be $P_1P_2P_{10}P_9P_8P_7$ and the second will be $P_{10}P_2P_3P_4P_{11}$. The modified perimeter will now be $P_{11}P_4P_5P_6P_7P_8P_9P_{10}$ and the next polygon to be found would be $P_{11}P_4P_5P_6P_8P_9$ but this splits the remaining polygon into two disjoint parts bounded by $P_6P_7P_8$ and $P_9P_{10}P_{11}$, respectively. These could be handled separately, but to avoid unnecessary program complications it is simpler to reject this polygon and recommence with P_6P_7 as initial line.

The process continues until all the points on the modified boundary perimeter lie in one plane which is also the nearest to the viewpoint.

The program can now either terminate, if simply a perspective view of a solid body is required, or else repeat itself with nearest planes and points replaced by farthest and with edges drawn either with solid or dotted lines depending on whether the polyhedron is transparent or semi-transparent.

Stereographic and isometric projections and illuminated polyhedra

These different types of projection are now relatively easily obtained using the basic method described above. To obtain stereographic effects as shown in Fig. 2 it is only necessary to repeat the calculation from an adjacent viewpoint so as to reproduce the effect of viewing with the two eyes. The actual viewpoints do not have to be very accurately chosen since the thickness of the lines drawn by the plotter masks any minor inaccuracies, and in any case normal vision is able to accommodate such inaccuracies.

Trimetric projections are similarly found by taking a viewpoint in the required direction but a long way away from the object. Again the naked eye is unable to distinguish the resulting drawing from a theoretical view from infinity. Isometric drawings can be obtained from a trimetric drawing taken from a suitably chosen point so as to make the projection of the cartesian axes equally inclined to one another. **Fig. 8** shows an isometric projection of a semi-transparent cube. The vertices of the cube were taken with unit coordinates and the viewpoint was at (100, 100, 100).

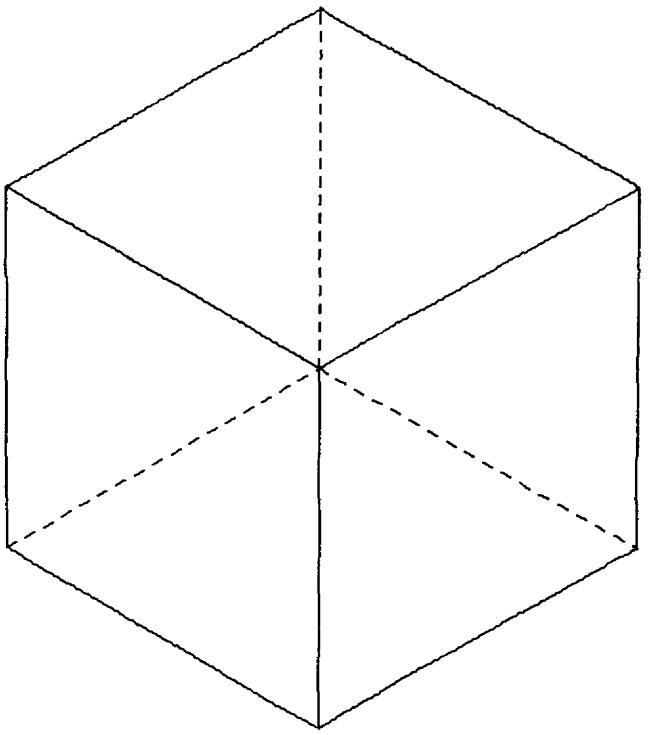


Fig. 8.—Isometric perspective view of a unit cube

Finally, illuminated polyhedra can be produced by shading the faces of a solid polyhedron as they are found by a uniform shading whose density is proportional to the angle between the outward-drawn normal to the current plane and the direction of the source of illumination. The drawing in Fig. 3 was shaded with a uniform shading of parallel lines, and the effect may possibly be improved by using some other form of shading. This possibility has not been investigated since the diagrams were all drawn with an off-line plotter on which even the simple shading shown uses a considerable amount of tape.

The diagrams reproduced here were all prepared using an Elliott 803 computer with a Benson-Lehner off-line plotter. I am indebted to Mr. B. Archer who wrote the subroutine to produce a uniform shading for a convex polygon.

Reference

GELLERT, GEORGE O. (1965). "Geometric Computing," *Machine Design*, Vol. 37, p. 95 and p. 152.

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